

CUSTOMER	:	RANBAXY
DESCRIPTION	:	Dibose 50mg
MEASUREMENT	:	W: 210MM X H: 177MM
COLOR	:	NBLU 0072

Date of revision : 1.11.19

If strongly distressing symptoms develop in spite of adherence to the diabetic diet prescribed, the doctor must be consulted and the dose temporarily or permanently reduced.

In patients receiving the recommended daily dose of 150 to 300 mg acarbose, clinically relevant abnormal liver function tests (three times above upper limit of normal range) were rarely observed. Abnormal values may be transient under ongoing acarbose therapy.

Other Abnormal Laboratory Findings: Small reductions in hematocrit but no association with reductions in hemoglobin have been reported in patients receiving acarbose. Low serum calcium and low plasma vitamin B6 levels were associated with acarbose therapy but are thought to be either spurious or of no clinical significance.

OVERDOSE

Unlike sulfonylureas or insulin, an overdose of acarbose will not result in hypoglycemia. An overdose may result in transient increases in flatulence, diarrhea, and abdominal discomfort, which shortly subside. In cases of overdosage the patient should not be given drinks or meals containing carbohydrates (polysaccharides, oligosaccharides and disaccharides) for the next 4-6 hours. Diarrhoea should be treated by standard conservative measures.

STORAGE

Store below 30°C, protected from moisture.

Do not expose to direct heat. The tablets should be peeled off from the foil just before use.

Discoloration of unpacked tablets can occur if exposed for more than 15 days at temperature higher than 25°C and Relative Humidity more than 60%. Dispensing of tablets in original pack is recommended.

KEEP ALL MEDICINES OUT OF REACH OF CHILDREN

SUPPLY

DIBOSE TABLETS 50 mg

Blister strips of 10, Box of 10 x 10's (Cold Form)

Date of revision : 31 Oct 2019

Product Registration Holder for Malaysia /
Manufactured by :

RANBAXY (MALAYSIA) SDN. BHD.

Lot 23, Bakar Arang Industrial Estate,
08000 Sungai Petani, Kedah, Malaysia.

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For the use only of a Registered Medical Practitioner

DIBOSE TABLETS (Acarbose Tablets 50mg)

COMPOSITION

DIBOSE TABLETS 50 mg

Each tablet contains:

Acarbose 50 mg

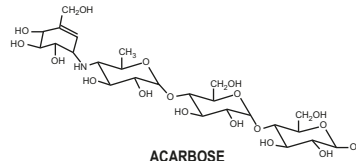
Inactive ingredients: Magnesium stearate, maize starch, microcrystalline cellulose, colloidal silicon dioxide

PRODUCT DESCRIPTION:

Off white to slightly yellow, uncoated, round, biconvex tablets with breakline on one side

DESCRIPTION

Acarbose is an oral alpha-glucosidase inhibitor for use in the management of type 2 Diabetes mellitus. Acarbose is an oligosaccharide, which is obtained from fermentation processes of a microorganism, *Actinoplanes utahensis*. It is chemically known as O-4, 6-dideoxy- 4-[[[(1S,4R,5S,6S)-4,5,6-trihydroxy-3-(hydroxymethyl)-2-cyclohexen-1-yl]amino]-α-D-glucopyranosyl-(1→4)-O-α-D-glucopyranosyl-(1→4)-D-glucose. Its empirical formula is C₂₈H₄₃NO₁₈ and its chemical structure is as follows:



PHARMACODYNAMIC AND PHARMACOKINETIC PROPERTIES

Mechanism of Action

The antihyperglycemic action of acarbose results from a competitive, reversible inhibition of pancreatic alpha-amylase and membrane-bound intestinal alpha-glucosidase hydrolase enzymes. Pancreatic alpha amylase hydrolyzes complex starches to oligosaccharides in the lumen of the small intestine, while the membrane-bound intestinal alpha-glucosidases hydrolyze oligosaccharides, trisaccharides, and disaccharides to glucose and other monosaccharides in the brush border of the small intestine. In diabetic patients, this enzyme inhibition results in a delayed glucose absorption and a lowering of postprandial hyperglycemia.

Because its mechanism of action is different, the effect of acarbose to enhance glycemic control is additive to that of sulfonylureas, insulin or metformin when used in combination. In addition, acarbose also diminishes the insulinotropic and weight-increasing effects of sulfonylureas. In contrast to sulfonylureas, it does not enhance insulin secretion.

Acarbose has no inhibitory activity against lactase and consequently would not be expected to induce lactose intolerance.

Pharmacokinetics

• Absorption:

It has been reported that in healthy men, less than 2% of an oral dose of acarbose was absorbed as active drug, while approximately 35% of total radioactivity from a 14C-labeled oral dose was absorbed. An average of 51% of an oral dose was excreted in the feces as unabsorbed drug-related radioactivity within 96 hours of ingestion. Because acarbose acts locally within the gastrointestinal tract, this low systemic bioavailability of parent compound is therapeutically desired. Following oral dosing with 14C-labeled acarbose, peak plasma concentrations of radioactivity were attained 14-24 hours after dosing, while peak plasma concentrations of active drug were attained at approximately 1 hour. The delayed absorption of acarbose-related radioactivity reflects the absorption of metabolites that may be formed by either intestinal bacteria or intestinal enzymatic hydrolysis.

• Metabolism:

Acarbose is metabolized exclusively within the gastrointestinal tract, principally by intestinal bacteria, but also by digestive enzymes. A fraction of these metabolites were absorbed and subsequently excreted in the urine. At least 13 metabolites have been separated. The major metabolites have been identified as 4- methylpyrogallol derivatives (i.e., sulfate, methyl, and glucuronide conjugates). One metabolite (formed by cleavage of a glucose molecule from acarbose) also has alpha-glucosidase inhibitory activity. This metabolite, together with the parent compound, accounts for less than 2% of the total administered dose.

• Excretion:

The fraction of acarbose that is absorbed as intact drug is almost completely excreted by kidneys. When acarbose was given intravenously, 89% of the dose was recovered in the urine as active drug within 48 hours. In contrast, less than 2% of an oral dose was recovered in the urine as active (i.e., parent compound and active metabolite) drug. This is consistent with the low bioavailability of the parent drug. The plasma elimination half-life of acarbose activity is approximately 2 hours in healthy volunteers. Consequently, drug accumulation does not occur with three times a day (i.i.d.) oral dosing.

• Special Populations:

The mean steady-state area under the curve (AUC) and maximum concentrations of acarbose were approximately 1.5 times higher in elderly compared to young volunteers; however, these differences were not statistically significant. Patients with severe renal impairment (Cl_{cr} < 25 mL/min/1.73m²) attained about 5 times higher peak plasma concentrations of acarbose and 6 times larger AUCs than volunteers with normal renal function. No studies of acarbose pharmacokinetic parameters according to race have been performed.

• Drug-Drug Interactions

Acarbose has no effect on either the pharmacokinetics or pharmacodynamics of nifedipine, propranolol, or ranitidine. Acarbose did not interfere with the absorption or disposition of the sulfonylurea glyburide in diabetic patients. Acarbose may affect digoxin bioavailability and may require dose adjustment of digoxin by 16%, decrease mean C_{max} of digoxin by 26% and decreases mean

trough concentrations of digoxin by 9%.

INDICATIONS

ACARBOSE TABLETS is recommended for the treatment of non-insulin dependent (NIDDM) diabetes mellitus in patients inadequately controlled on diet alone, or on diet and oral hypoglycaemic agents

DOSE AND METHOD OF ADMINISTRATION

ACARBOSE TABLETS are taken orally and should be chewed with the first mouthful of food, or swallowed whole with a little liquid directly before the meal. Owing to the great individual variation of glucosidase activity in the intestinal mucosa, there is no fixed dosage regimen, and patients should be treated according to clinical response and tolerance of intestinal side-effects.

Adults

The recommended initial dose is 50 mg three times a day. However, some patients may benefit from more gradual initial dose titration to minimise gastrointestinal side-effects. This may be achieved by initiating treatment at 50 mg once or twice a day, with subsequent titration to a three times a day regimen.

If after six to eight weeks' treatment patients show an inadequate clinical response, the dosage may be increased to 100 mg three times a day. A further increase in dosage to a maximum of 200 mg three times a day may occasionally be necessary.

If distressing complaints develop in spite of strict adherence to the diet, the dose should not be increased further and if necessary should be reduced according to the severity of the side-effects and the clinical judgment of the prescriber.

Acarbose is intended for continuous long-term treatment.

Elderly patients

No modification of the normal adult dosage regimen is necessary.

Children and adolescents under 18 years

The efficacy and safety of acarbose in children and adolescents have not been established. Acarbose is not recommended for patients under the age of 18 years.

CONTRAINDICATIONS

- patients with known hypersensitivity to the acarbose or any of the excipients
- patients with inflammatory bowel disease, colonic ulceration, and partial intestinal obstruction or in patients predisposed to intestinal obstruction
- patients who have chronic intestinal diseases associated with marked disorders of digestion or absorption and in patients who have conditions that may deteriorate as a result of increased gas formation in the intestine, e.g. larger hernias, Roemheld syndrome
- pregnancy and in nursing mothers
- patients with severe hepatic impairment
- patients with diabetic ketoacidosis or cirrhosis
- As acarbose has not been studied in patients with severe renal impairment, it should not be used in patients with a creatinine clearance < 25 ml/min/1.73m².

WARNINGS AND PRECAUTIONS

Hypoglycaemia: When administered alone, acarbose does not cause hypoglycaemia. It may, however, act to potentiate the hypoglycaemic effects of insulin, metformin and sulphonylurea drugs, and the dosages of these agents may need to be modified accordingly. If acute hypoglycemia develops glucose should be used for rapid correction of hypoglycaemia.

Episodes of hypoglycaemia occurring during therapy must, where appropriate, be treated by the administration of glucose, not sucrose. This is because acarbose will delay the digestion and absorption of disaccharides, but not monosaccharides.

Transaminases: Patients treated with acarbose may, on rare occasions, experience an idiosyncratic response with either symptomatic or asymptomatic hepatic dysfunction. In the majority of cases this dysfunction is reversible on discontinuation of acarbose therapy. It is recommended that liver enzyme monitoring is considered during the first six to twelve months of treatment. If elevated transaminases are observed, withdrawal of therapy may be warranted, particularly if the elevations persist. In such circumstances, patients should be monitored at weekly intervals until normal values are established.

The administration of antacid preparations containing magnesium and aluminium salts, e.g. hydrotalcite, has been shown not to ameliorate the acute gastrointestinal symptoms of acarbose in higher dosage and should, therefore, not be recommended to patients for this purpose.

Macrovascular Outcomes: No conclusive clinical evidence of macrovascular risk reduction has been reported with acarbose or any other anti-diabetic drug.

Loss of Control of Blood Glucose: When diabetic patients are exposed to stress such as fever, trauma, infection, or surgery, a temporary loss of control of blood glucose may occur. At such times, temporary insulin therapy may be necessary.

Laboratory Tests

Therapeutic response to acarbose should be monitored by periodic blood glucose tests. Measurement of glycosylated hemoglobin levels is recommended for the monitoring of long-term glycemic control.

Acarbose, particularly at doses in excess of 50 mg t.i.d., may give rise to elevations of serum transaminases and, in rare instances, hyperbilirubinemia. It is recommended that serum transaminase levels be checked every 3 months during the first year of treatment with acarbose and periodically thereafter. If elevated transaminases are observed, a reduction in dosage or withdrawal of therapy may be indicated, particularly if the elevations persist.

Effects on ability to drive and use machines

None known.

USE IN SPECIAL POPULATIONS

Pregnancy and Lactation

Acarbose should not be administered during pregnancy as no information is available on its use in pregnant women (see **CONTRAINDICATIONS**).

It has been reported that a small amount of acarbose or its metabolites was recovered in the milk of nursing rats. To date there have been no similar findings in humans.

Nevertheless, as the possibility of drug induced effects on nursing infants cannot be excluded, the prescription of acarbose is not recommended during breastfeeding (see **CONTRAINDICATIONS**).

DRUG INTERACTIONS

Certain drugs tend to produce hyperglycemia and may lead to loss of blood glucose control. These drugs include the thiazides and other diuretics, corticosteroids, phenothiazines, thyroid products, estrogens, oral contraceptives, phenytoin, nicotinic acid, sympathomimetics, calcium channel blocking drugs, and isoniazid. When such drugs are administered to a patient receiving acarbose, the patient should be closely observed for loss of blood glucose control. When such drugs are withdrawn from patients receiving acarbose in combination with sulfonylureas or insulin, patients should be observed closely for any evidence of hypoglycemia.

Sulfonylurea agents or insulin may cause hypoglycemia. Acarbose given in combination with a sulfonylurea, metformin or insulin may cause a further lowering of blood glucose and may increase the potential for hypoglycemia. If hypoglycemia occurs, appropriate adjustments in the dosage of these agents should be made. Very rarely, individual cases of hypoglycemic shock have been reported in patients receiving acarbose therapy in combination with sulfonylureas and/or insulin.

Episodes of hypoglycaemia occurring during therapy must, where appropriate, be treated by the administration of glucose, not sucrose. This is because acarbose will delay the digestion and absorption of disaccharides, but not monosaccharides.

Sucrose (cane sugar) and foods containing sucrose often cause abdominal discomfort or even diarrhoea during treatment with acarbose as a result of increased carbohydrate fermentation in the colon.

Intestinal adsorbents (e.g., charcoal) and digestive enzyme preparations containing carbohydrate-splitting enzymes (e.g., amylase, pancreatin) may reduce the effect of acarbose and should not be taken concomitantly.

Acarbose has been shown to change the bioavailability of digoxin when they are coadministered, which may require digoxin dose adjustment. Monitoring of serum digoxin levels should be considered.

The concomitant administration of neomycin may lead to enhance reductions of postprandial blood glucose and to an increase in the frequency and severity of gastro-intestinal side-effects. If the symptoms are severe, a temporary dose reduction of acarbose may be warranted.

The concomitant administration of colestyramine may enhance the effects of acarbose, particularly with respect to reducing postprandial insulin levels. Simultaneous administration of acarbose and colestyramine should, therefore, be avoided. In the rare circumstance that both acarbose and colestyramine therapy are withdrawn simultaneously, care is needed as a rebound phenomenon has been observed with respect to insulin levels in non-diabetic subjects.

No significant or reproducible changes in the plasma nifedipine profiles have been reported with concomitant administration of acarbose and nifedipine.

SIDE EFFECTS/ADVERSE REACTIONS

The adverse drug reactions (ADRs) reported with acarbose are summarised in the table below.

System Organ Class (MedDRA)	Very common (≥1/10)	Common (≥1/100 to < 1/10)	Uncommon (≥1/1,000 to < 1/100)	Rare (≥1/10,000 to < 1/1,000)	Not known (frequency could not be established)
Blood and lymphatic system disorders					Thrombocytopenia
Immune system disorders					Allergic reaction (rash, erythema, exanthema, urticaria)
Vascular disorders				Oedema	
Gastrointestinal disorders	Flatulence	Diarrhoea Gastrointestinal and abdominal pains	Nausea Vomiting Dyspepsia		Subileus/ileus Pneumatosis cystoides intestinalis
Hepatobiliary disorders			Increase in liver enzymes	Jaundice	Hepatitis

In addition, liver disorder, hepatic function abnormal and liver injury have also been reported in Japan. Individual cases of fulminant hepatitis with fatal outcome have been reported in Japan. The relationship to acarbose is unclear.

If the prescribed diabetic diet is not observed the intestinal side effects may be intensified.